

PHYS 486 - Problem Set 1

1.a)

$$\rightarrow \nu \lambda = c \Rightarrow \nu_1 = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ m/s}}{700 \text{ nm}} \approx 4.29 \times 10^{14} \text{ Hz} \Rightarrow E_1 = h \nu_1 = (6.63 \times 10^{-27} \text{ erg.s}) (4.29 \times 10^{14} \text{ Hz}) \\ = 2.84 \times 10^{-12} \text{ erg}$$
$$\nu_2 = \frac{3 \times 10^8}{400 \text{ nm}} = 7.5 \times 10^{14} \text{ Hz} \Rightarrow E_2 = h \nu_2 \approx 4.97 \times 10^{-12} \text{ erg}$$

Frequency is between 429 to 750 THz and energy is btwn. 2.84×10^{-12} to 4.97×10^{-12} erg

$$\rightarrow \text{Microwave: } P = 7.5 \times 10^9 \text{ erg} = \frac{\# \text{ photons}}{\text{s}} \times h \nu \Rightarrow \frac{\# \text{ photons}}{\text{second}} = \frac{7.5 \times 10^9 \text{ erg/s}}{6.63 \times 10^{-27} \text{ erg.s} \times 2.5 \times 10^9 \frac{1}{\text{s}}} \approx 4.52 \times 10^{26} \frac{\text{photon}}{\text{s}}$$

$$\text{Laser: } \nu = \frac{3 \times 10^8}{633 \times 10^{-9}} = 4.74 \times 10^{14} \text{ Hz} \Rightarrow \frac{\# \text{ photons}}{\text{s}} = \frac{10^4 \text{ erg/s}}{6.63 \times 10^{-27} \text{ erg.s} \times 4.74 \times 10^{14} \text{ Hz}} \approx 3.18 \times 10^{15} \text{ photons/s}$$

$$\text{Cell phone: } \frac{\# \text{ photons}}{\text{second}} = \frac{4 \times 10^6 \text{ erg/s}}{6.63 \times 10^{-27} \text{ erg.s} \times 850 \times 10^6 \text{ Hz}} = 7.10 \times 10^{23} \text{ photons/s}$$

$\rightarrow$ At a given power, radio waves with lower freq. (and thus lower energy) have much more photons emitted compared to X-rays. This higher density photon current makes the beam appear as a continuous stream rather than quantized energy packets (photons).

Thus radio waves behave more classically

1.b) $\rightarrow$ Car : $p = 2000 \text{ kg} \times 16 \text{ m/s} = 32\,000 \text{ kg m/s} \Rightarrow \lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34} \text{ m}^2 \text{ kg/s}}{3.2 \times 10^4 \text{ m kg/s}} = \boxed{2.07 \times 10^{-38} \text{ m}}$
extremely small

$\rightarrow$ Marble : $p = 10 \text{ g} \times 10 \text{ cm/s} = 100 \text{ g cm/s} \Rightarrow \lambda = \frac{h}{p} = \frac{6.63 \times 10^{-27} \text{ cm}^2 \text{ g/s}}{100 \text{ cm g/s}} = \boxed{6.63 \times 10^{-31} \text{ m}}$

$\rightarrow$ Smoke particle : $m = \frac{4}{3} \pi \left(\frac{10^{-5} \text{ cm}}{2} \right)^3 \times 0.2 \text{ g/cm}^3 \simeq 1.05 \times 10^{-19} \text{ g}$

$KE = \frac{p^2}{2m} = \frac{3}{2} k_B T \Rightarrow p = \sqrt{3 k_B T m} = \sqrt{3 \times 1.38 \times 10^{-16} \text{ erg} \times 300 \text{ K} \times 1.05 \times 10^{-19} \text{ g}} \simeq 3.606 \times 10^{-15} \text{ g cm/s}$

$\lambda = \frac{6.63 \times 10^{-27} \text{ cm}^2 \text{ g/s}}{3.606 \times 10^{-15} \text{ cm g/s}} = \boxed{1.84 \times 10^{-14} \text{ m}}$

$\rightarrow$ ^{87}Rb $p = \sqrt{3 k_B T m} = \sqrt{3 \times 1.38 \times 10^{-16} \text{ erg} \times 10^{-4} \text{ K} \times 1.44 \times 10^{-22} \text{ g}} \simeq 2.446 \times 10^{-21} \text{ g cm/s}$

$\lambda = \frac{6.63 \times 10^{-27} \text{ erg s}}{2.446 \times 10^{-21} \text{ g cm/s}} \simeq 2.71 \times 10^{-8} \text{ m} = \boxed{27.1 \text{ nm}}$

$$2.) \quad \Psi(x,t) = A \frac{1}{\lambda x^2 + 1} e^{-i\omega t}$$

a) Normalization requires:

$$\text{Let } x = \frac{1}{\sqrt{\lambda}} \tan \theta \Rightarrow dx = \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}} \quad \left| \begin{array}{l} x \rightarrow \infty \Rightarrow \theta \rightarrow \frac{\pi}{2} \\ x \rightarrow -\infty \Rightarrow \theta \rightarrow -\frac{\pi}{2} \end{array} \right.$$

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} dx |\Psi(x,t)|^2 = A^2 \int_{-\infty}^{\infty} dx \frac{1}{(\lambda x^2 + 1)^2} = A^2 \int_{-\pi/2}^{\pi/2} \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}} \frac{1}{(\underbrace{\tan^2 \theta + 1}_{\sec^2 \theta})^2} = \frac{A^2}{\sqrt{\lambda}} \int_{-\pi/2}^{\pi/2} d\theta \cos^2 \theta \\ &= \frac{A^2}{\sqrt{\lambda}} \int_{-\pi/2}^{\pi/2} d\theta \left(\frac{1}{2} + \frac{1}{2} \cos 2\theta \right) = \frac{A^2}{2\sqrt{\lambda}} \left[\theta + \frac{\sin 2\theta}{2} \right]_{-\pi/2}^{\pi/2} = \frac{A^2}{2\sqrt{\lambda}} (\pi + 0) \end{aligned}$$

$$\Rightarrow \boxed{A = \sqrt{\frac{2\sqrt{\lambda}}{\pi}}}$$

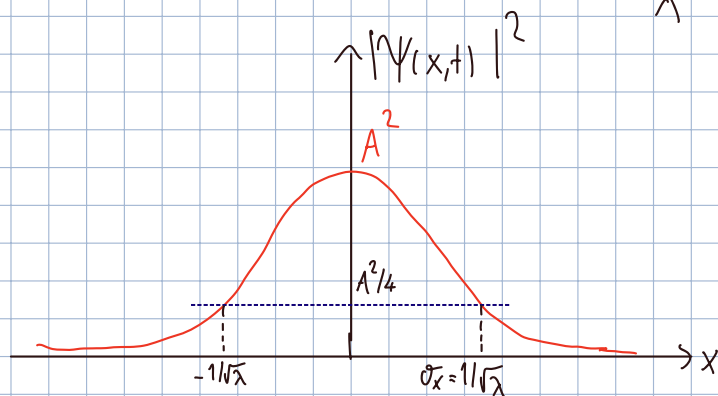
$$\langle x \rangle = \int_{-\infty}^{\infty} dx x |\Psi(x,t)|^2 = A^2 \int_{-\infty}^{\infty} dx \frac{x}{(\lambda x^2 + 1)^2} = 0 \quad * \text{ Notice that integrand is an odd function and limits are symmetric. Thus, integral is zero.}$$

$$\boxed{\langle x \rangle = 0}$$

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} dx x^2 |\Psi(x,t)|^2 = A^2 \int_{-\infty}^{\infty} dx \frac{x^2}{(\lambda x^2 + 1)^2} \quad \text{Let } x = \frac{\tan \theta}{\sqrt{\lambda}} \Rightarrow dx = \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}} \quad \text{Limits are } -\pi/2, +\pi/2 \text{ again}$$

$$\begin{aligned} \langle x^2 \rangle &= A^2 \int_{-\pi/2}^{\pi/2} \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}} \cdot \frac{\tan^2 \theta}{\lambda} \frac{1}{(\underbrace{\tan^2 \theta + 1}_{\sec^2 \theta})^2} = \frac{A^2}{\lambda \sqrt{\lambda}} \int_{-\pi/2}^{\pi/2} \sin^2 \theta d\theta \\ &= \frac{A^2}{2\lambda \sqrt{\lambda}} \int_{-\pi/2}^{\pi/2} (1 - \cos 2\theta) d\theta = \frac{A^2 \pi}{2\lambda \sqrt{\lambda}} = \frac{2\sqrt{\lambda}}{\pi} \frac{\pi}{2\lambda \sqrt{\lambda}} = \boxed{\frac{1}{\lambda}} \end{aligned}$$

$$2.b) \quad \sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{\lambda} - 0 \Rightarrow \sigma_x = \frac{1}{\sqrt{\lambda}}$$



$$|\Psi(x,t)|^2 = \frac{A^2}{(\lambda x^2 + 1)^2} \Rightarrow \text{stationary}$$

$$|\Psi(x = \frac{1}{\sqrt{\lambda}}, t)|^2 = \frac{A^2}{4}, \quad |\Psi(x=0)|^2 = A^2$$

$$\Pr\{\langle x \rangle - \sigma \leq x \leq \langle x \rangle + \sigma\} = 2 \times \Pr\{0 \leq x \leq \frac{1}{\sqrt{\lambda}}\} \Rightarrow \text{due to evenness of } |\Psi|^2$$

$$\Pr\{0 \leq x \leq \frac{1}{\sqrt{\lambda}}\} = A^2 \int_0^{1/\sqrt{\lambda}} \frac{dx}{(\lambda x^2 + 1)^2} \Rightarrow \text{Let } x = \frac{\tan \theta}{\sqrt{\lambda}}, \quad dx = \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}}$$

Limits are from $\theta=0$ to $\theta = \tan^{-1}\{1\} = \frac{\pi}{4}$

From previous options we've derived that

$$= \frac{A^2}{2\sqrt{\lambda}} \int_0^{\pi/4} (1 + \cos 2\theta) d\theta = \frac{A^2}{2\sqrt{\lambda}} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/4} = \frac{1}{\pi} \left(\frac{\pi}{4} + \frac{1}{2} \right) = \frac{1}{4} + \frac{1}{2\pi}$$

$$\Pr\{x > \sigma \cup x < -\sigma\} = 1 - 2 \Pr\{0 \leq x \leq \sigma\} = 1 - \frac{1}{2} - \frac{1}{\pi} = \frac{1}{2} - \frac{1}{\pi} = \boxed{18.2\%}$$

$$2.c) \quad \hat{p} = -i\hbar \frac{d}{dx}, \quad \hat{p} \Psi(x,t) = +i\hbar A \frac{2\lambda x}{(\lambda x^2 + 1)^2} e^{-i\omega t} = i\hbar A \frac{2\lambda x}{\lambda x^2 + 1} \Psi(x,t)$$

$$\langle p \rangle = \int_{-\infty}^{\infty} dx \Psi^*(x,t) \underbrace{(-i\hbar) \frac{\partial \Psi(x,t)}{\partial x}}_{\hat{p} \Psi} = 2\lambda i\hbar A \int_{-\infty}^{\infty} dx \frac{x}{\lambda x^2 + 1} \underbrace{\Psi^*(x,t) \Psi(x,t)}_{|\Psi(x,t)|^2} \Rightarrow \text{Notice that } \frac{x}{\lambda x^2 + 1} \text{ is an odd function and } |\Psi|^2 \text{ is even. Thus integrand is odd. Since limits are symmetric integral is zero}$$

$$\boxed{\langle p \rangle = 0}$$

$$\langle p^2 \rangle = \int_{-\infty}^{\infty} dx \Psi^*(x,t) \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \right) \Psi(x,t) = -\hbar^2 \int_{-\infty}^{\infty} dx \Psi^* \frac{\partial^2 \Psi}{\partial x^2}$$

$$= -\hbar^2 \left[\cancel{\Psi^* \frac{\partial \Psi}{\partial x}} \right]_{-\infty}^{\infty} + \hbar^2 \int_{-\infty}^{\infty} dx \left| \frac{\partial \Psi}{\partial x} \right|^2$$

0 since $\lim_{x \rightarrow \pm \infty} \Psi = 0$

$$\left| \frac{\partial \Psi}{\partial x} \right|^2 = 4\hbar^2 \lambda^2 A^2 \frac{x^2}{(\lambda x^2 + 1)^4}$$

using the result from previous option

Use integration by parts

$$du = \frac{\partial^2 \Psi}{\partial x^2}, \quad v = \Psi^*$$

$$u = \frac{\partial \Psi}{\partial x}, \quad dv = \frac{\partial \Psi^*}{\partial x} dx$$

$$\langle p^2 \rangle = 4\hbar^2 \lambda^2 A^2 \int_{-\infty}^{\infty} dx \frac{x^2}{(\lambda x^2 + 1)^4} \quad \text{Letting } x = \frac{\tan \theta}{\sqrt{\lambda}} \quad \text{once more } dx = \frac{\sec^2 \theta d\theta}{\sqrt{\lambda}}$$

$$= 4\hbar^2 \lambda^2 A^2 \int_{-\pi/2}^{\pi/2} d\theta \frac{1}{\lambda} \frac{\sec^2 \theta \tan^2 \theta / \lambda}{\sec^8 \theta} = 4\hbar^2 \frac{2\sqrt{\lambda}}{\frac{\pi}{A^2}} \int_{-\pi/2}^{\pi/2} d\theta (\sin^2 \theta \cos^4 \theta) \quad \downarrow 1 - \cos^2 \theta$$

$$= \frac{8\hbar^2 \sqrt{\lambda}}{\pi} \left[\int_{-\pi/2}^{\pi/2} d\theta \cos^4 \theta - \int_{-\pi/2}^{\pi/2} d\theta \cos^6 \theta \right]$$

$$\int_{-\pi/2}^{\pi/2} d\theta \left(\frac{1 + \cos 2\theta}{2} \right)^2 = \int_{-\pi/2}^{\pi/2} d\theta \left[\frac{1}{4} + \frac{\cos 2\theta}{2} + \frac{1}{8} + \frac{\cos 4\theta}{8} \right] \quad \left| \quad \int_{-\pi/2}^{\pi/2} \underbrace{\cos^5 \theta}_{u} \underbrace{\cos \theta}_{dv} d\theta = \left[\cos^5 \theta \sin \theta \right]_{-\pi/2}^{\pi/2} + \int_{-\pi/2}^{\pi/2} 5 \cos^4 \theta \sin^2 \theta d\theta \right]$$

$$= \left[\frac{\theta}{4} + \frac{\sin 2\theta}{2} + \frac{\theta}{8} + \frac{\sin 4\theta}{32} \right]_{-\pi/2}^{\pi/2} = \frac{\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$$

$$I = 5 \int_{-\pi/2}^{\pi/2} d\theta \cos^4 \theta - 5 \int_{-\pi/2}^{\pi/2} d\theta \cos^6 \theta$$

$$\hookrightarrow I = \frac{5}{6} \int_{-\pi/2}^{\pi/2} d\theta \cos^4 \theta = \frac{5}{6} \times \frac{3\pi}{8} = \frac{5\pi}{16}$$

$$\langle p^2 \rangle = \frac{8\hbar^2 \sqrt{\lambda}}{\pi} \left(\frac{6\pi}{16} - \frac{5\pi}{16} \right) = \boxed{\frac{\hbar^2 \sqrt{\lambda}}{2}}$$

2.d)

$$\sigma_p^2 = \langle p^2 \rangle - \langle p \rangle^2 = \frac{\hbar^2 \sqrt{\lambda}}{2} \Rightarrow$$

$$\sigma_p = \hbar \sqrt{\frac{\sqrt{\lambda}}{2}} \Rightarrow \lambda = 0 \text{ minimizes } \sigma_p$$

$\hookrightarrow$ Wave function spread out in space.

$\hookrightarrow$ Approaches to a plane wave with a single wavenum. k

In contrast, $\lambda \rightarrow \infty$ minimizes σ_x

$\hookrightarrow$ Wave function localized in a single point

$$3) \quad \frac{d\langle p \rangle}{dt} = \frac{d}{dt} \int_{-\infty}^{\infty} dx \, \psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \psi = -i\hbar \int_{-\infty}^{\infty} dx \, \frac{\partial}{\partial t} \left[\psi^* \frac{\partial \psi}{\partial x} \right] = -i\hbar \int_{-\infty}^{\infty} dx \left[\frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial x} + \psi^* \frac{\partial^2 \psi}{\partial t \partial x} \right]$$

change the order of derivatives

$$= - \int_{-\infty}^{\infty} dx \, \psi^* \frac{\partial}{\partial x} \left[i\hbar \frac{\partial \psi}{\partial t} \right] - \int_{-\infty}^{\infty} dx \, i\hbar \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial x}$$

$-i\hbar \frac{\partial \psi^*}{\partial t} = i\hbar \frac{\partial \psi}{\partial t}$

$$= - \int_{-\infty}^{\infty} dx \, \psi^* \frac{\partial}{\partial x} \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi(x) \right] + \int_{-\infty}^{\infty} dx \left[-\frac{\hbar^2}{2m} \frac{\partial^2 \psi^*}{\partial x^2} + V(x) \psi^*(x) \right] \frac{\partial \psi}{\partial x}$$

$$= + \frac{\hbar^2}{2m} \underbrace{\int_{-\infty}^{\infty} dx \, \psi^* \frac{\partial^3 \psi}{\partial x^3}}_{\substack{u \\ dv}} - \int_{-\infty}^{\infty} dx \, \psi^* \frac{\partial V(x)}{\partial x} \psi - \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} dx \, \frac{\partial^2 \psi^*}{\partial x^2} \frac{\partial \psi}{\partial x} = - \left\langle \frac{\partial V(x)}{\partial x} \right\rangle$$

$$\left[\cancel{\psi^* \frac{\partial^2 \psi}{\partial x^2}} \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} dx \, \underbrace{\frac{\partial \psi^*}{\partial x}}_u \underbrace{\frac{\partial^2 \psi}{\partial x^2}}_{dv}$$

0 since $\lim_{x \rightarrow \pm\infty} |\psi| = 0$

$$= - \left[\cancel{\frac{\partial \psi^*}{\partial x} \frac{\partial \psi}{\partial x}} \right] + \int_{-\infty}^{\infty} dx \, \frac{\partial^2 \psi^*}{\partial x^2} \frac{\partial \psi}{\partial x} //$$

0 since first derivatives must vanish at infinity for wavefunc. to be normalizable

⇒ cancels out with

↳ We have used integration by parts twice

Ehrenfest
Theorem

$$\frac{d\langle p \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle$$

⇒ In conformity with the classical view:

$$\frac{dp}{dt} = F = - \nabla V(x) //$$

4) Using T.I. S.E.

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \Psi'}{\partial x^2} + \underbrace{V(x)}_{V(x)+V_0} \Psi' = \frac{\hbar^2}{2m} \Psi' + V(x) \Psi' + V_0 \Psi' \quad + i\hbar \frac{\partial \Psi'}{\partial t} = E' \Psi'$$

$$\hookrightarrow (E' - V_0) \Psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V(x) \Psi(x) = E \Psi(x) \Rightarrow E' = E + V_0$$

Essentially every state's energy is shifted up to $E + V_0$.

$$\hookrightarrow \text{For a stationary state } \Psi'(x,t) = \Psi(x,t) e^{-iV_0 t/\hbar} = \Psi(x) e^{-i(E+V_0)t/\hbar}$$

In the case of time independent Hamiltonian, solutions of S.E. will be stationary

$$\rightarrow \text{Notice } |\Psi'(x,t)|^2 = |\Psi(x)|^2 = |\Psi(x,t)|^2 \Rightarrow \text{complex exponential cancel}$$

$$\text{Thus } \langle x \rangle' = \int_{-\infty}^{\infty} dx \Psi'(x,t) x \Psi'(x,t) = \int_{-\infty}^{\infty} dx x |\Psi'(x,t)|^2 = \int_{-\infty}^{\infty} dx x |\Psi(x,t)|^2 = \langle x \rangle$$

$\hookrightarrow$ Average position is the same as original potential case

In general, for any observable $\hat{O}$:

$$\langle \hat{O} \rangle' = \int dx \Psi_{(x,t)}^* e^{+iV_0 t/\hbar} \hat{O} \Psi_{(x,t)} e^{-iV_0 t/\hbar} = \int dx \Psi_{(x,t)}^* \hat{O} \Psi_{(x,t)} = \langle \hat{O} \rangle$$

$\hookrightarrow$ Phase factors cancel since $\hat{O}$ is time-indep and $e^{\pm iV_0 t/\hbar}$ terms can be reordered together.

V_0 potential only adds a global phase $\frac{V_0}{\hbar}$ to the wavefunction.

Measurable quantities will not change

It's probability distribution will remain the same, as well.